Practice workbook · Plot a line graph from a table of results

Continuous data in a table are plotted on a grid, for one or several series, joined with ruled lines or a smooth curve as instructed.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-PLOT-LINE-GRAPH

1. Question 2 · part a.i · 6 marks

November 2023 Paper 1

Worked-example retry · Attempt part a.i only.

2 To find out if a person has diabetes, they can be given a glucose tolerance test.

This is how the test is done.

- the person does not eat for 12 hours
- after the 12 hours, blood glucose levels are measured
- the person is then given a solution of glucose to drink
- the blood glucose levels of the person are measured every 30 minutes for 3 hours

The table shows the results for a person without diabetes and a person with diabetes.

Time in minutes	Blood glucose levels in mg per 100 cm^3	
	person without diabetes	person with diabetes
0	65	110
30	120	215
60	130	230
90	95	200
120	80	190
150	65	180
180	65	170

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1. Question 2 · part a.i · 6 marks

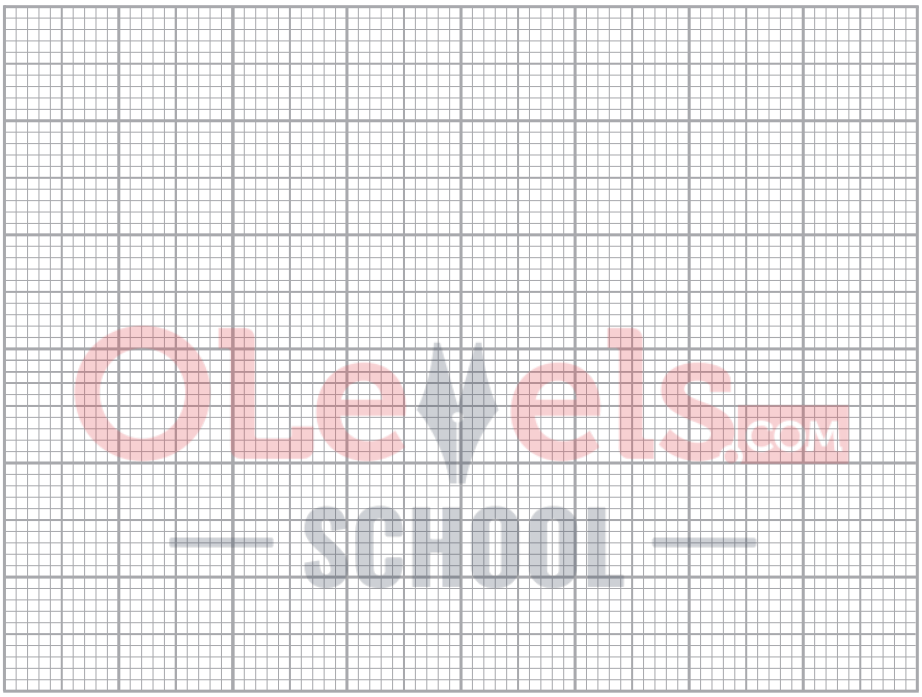
November 2023 Paper 1

Worked-example retry · Attempt part a.i only.

(a) (i) Plot the results on the grid.

Join the points with straight lines.

(6)



(ii) Determine the difference between the highest and lowest blood glucose levels for each person.

(2)

without diabetes = ... mg per 100 cm^3

with diabetes = ... mg per 100 cm^3

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1. Question 2 · part a.i · 6 marks · Working and answer

Working and answer

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2. Question 5 · part a · 5 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

5 A student investigates the effect of temperature on the volume of sweat produced.

This is the student's method.

- sit a person in a room at 20 °C
- measure the volume of sweat the person produces
- repeat with the room at different temperatures
- repeat again when the person is active

The table shows the student's results.

Temperature of room in °C	Volume of sweat produced in arbitrary units	
	Sitting	Active
20	1	2
22	2	6
24	2	8
26	2	11
28	3	14
30	4	22

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2. Question 5 · part a · 5 marks

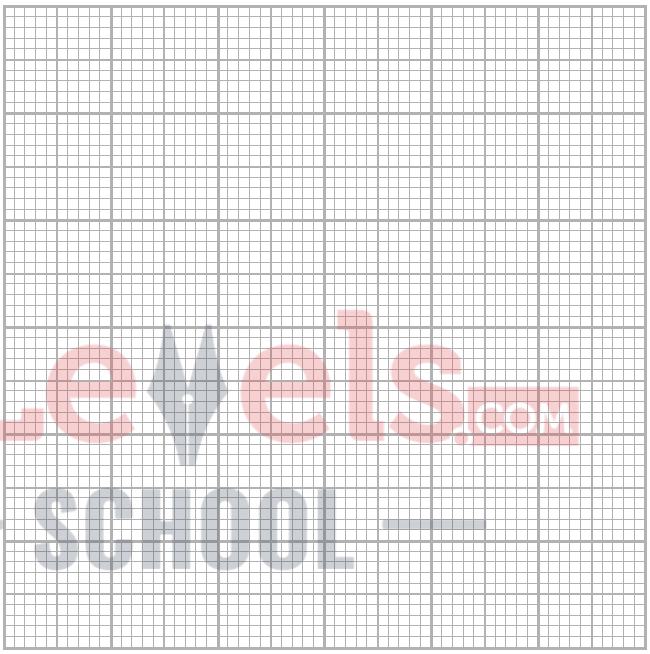
June 2019 Paper 1R

Further practice · Attempt part a only.

(a) Plot the student's results on the grid.

Join the points with straight lines.

(5)



(b) Use your graph to determine the difference in the volume of sweat produced at 27 °C when sitting and when active.

(2)

difference = ... arbitrary units

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2. Question 5 · part a · 5 marks · Working and answer

Working and answer

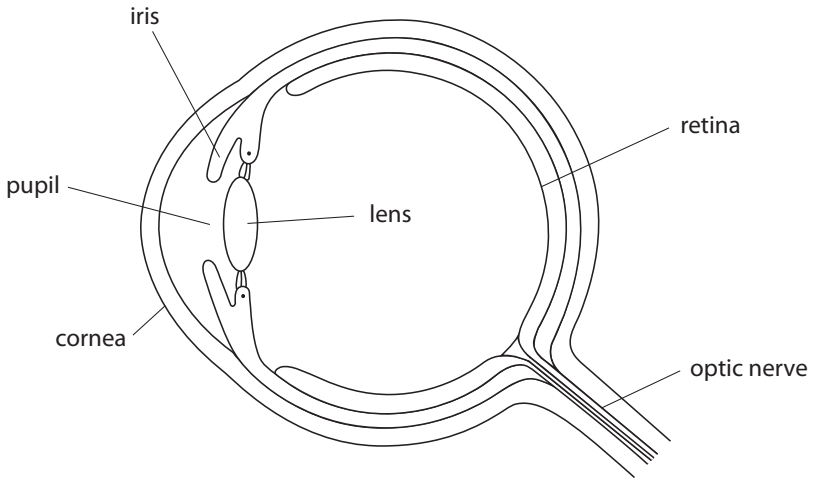
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3. Question 3 · part b.i · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.i only.

3 The human eye contains structures that help to focus light on the retina.



(a) Which structures help to focus light on the retina?

(1)

- ☒ **A** cornea and lens
- ☒ **B** iris and cornea
- ☒ **C** iris and pupil
- ☒ **D** pupil and lens

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3. Question 3 · part b.i · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.i only.

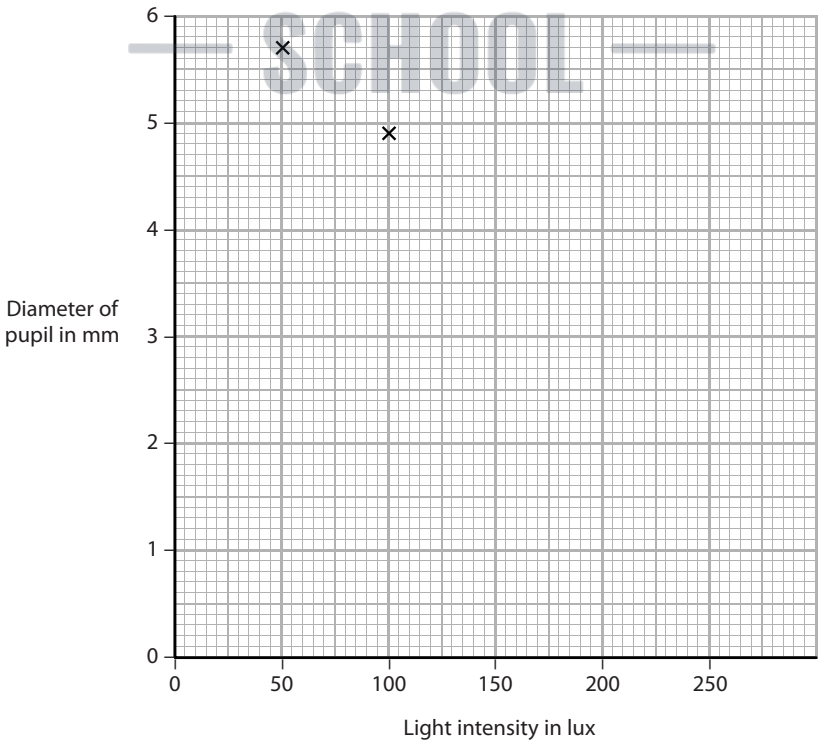
(b) The intensity of light entering the eye can be measured in a unit called lux.

The table shows the results of an investigation on how light intensity affects the diameter of the pupil in the eye.

Light intensity in lux	Diameter of pupil in mm
50	5.7
100	4.9
150	2.7
200	3.2
250	2.3

(i) Complete the graph by plotting the results for 150, 200 and 250 lux. (1)

(ii) Draw a straight line of best fit. (1)



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3. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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4. Question 5 · part a.i · 6 marks

November 2020 Paper 2R

Further practice · Attempt part a.i only.

5 A scientist investigates how the drug, atropine, affects the heart rate in humans.

Atropine paralyses the vagus nerve which carries impulses from the brain to the heart.

The heart rates of ten people are measured at intervals of two minutes for eight minutes, after which, atropine is injected.

The scientist continues to measure the heart rates.

The scientist's results are shown in the table.

Time in minutes	Heart rate in beats per minute
0	90
2	81
4	78
6	80
8	82
atropine injected → 9	100
10	120
11	148
12	160
13	138
15	128
17	120
19	116
21	109
23	100

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4. Question 5 · part a.i · 6 marks

November 2020 Paper 2R

Further practice · Attempt part a.i only.

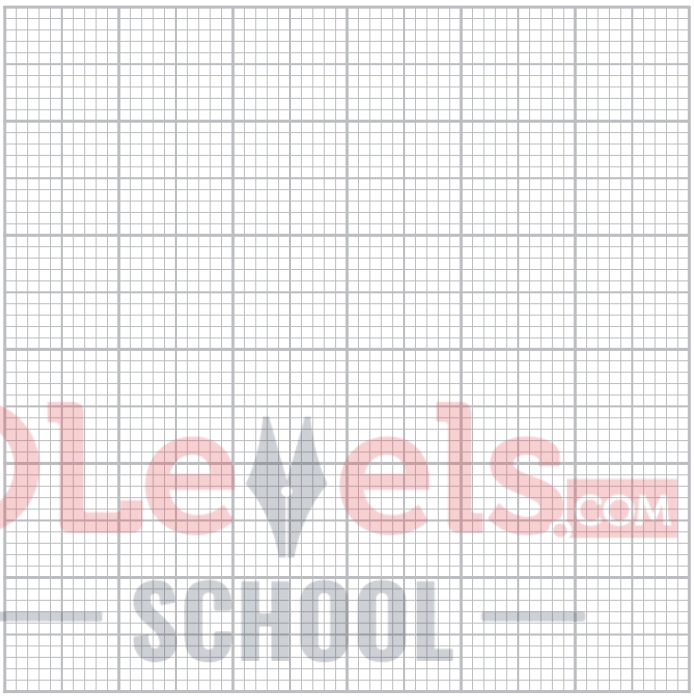
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(a) (i) Plot the scientist's results on the grid. Draw a curve of best fit.

(6)



(ii) Explain why ten people are injected with atropine rather than just one person.

(2)

.....

.....

.....

.....

(b) (i) Calculate the percentage change in the heart rate at 12 minutes compared to the heart rate at the start of the investigation.

(2)

percentage change = ...%

4. Question 5 · part a.i · 6 marks · Working and answer

Working and answer

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5. Question 4 · part a.i · 5 marks

November 2020 Paper 2

Further practice · Attempt part a.i only.

4 Aspirin is the active ingredient used in some tablets to relieve pain.

Aspirin becomes acidic when it dissolves in the body. The faster aspirin dissolves, the faster it can relieve pain.

A student investigates if it is better to mix sucrose or starch with the active ingredient to make an aspirin tablet.

This is the student's method.

- add distilled water to three beakers, A, B and C
- add 1 aspirin tablet to each beaker
- add 5 g of sucrose to beaker B
- add 5 g of starch to beaker C
- measure the pH of the contents of each beaker at the start of the investigation
- measure the pH of the contents of each beaker every minute for 5 minutes

The table shows the student's results.

Beaker	pH of contents of beaker					
	at start	1 minute	2 minutes	3 minutes	4 minutes	5 minutes
A – aspirin	7.6	7.1	6.6	6.2	5.8	5.4
B – aspirin and sucrose	7.2	6.7	6.3	5.7	5.2	4.7
C – aspirin and starch	7.1	7.2	7.3	7.3	7.4	7.4

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5. Question 4 · part a.i · 5 marks

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Further practice · Attempt part a.i only.

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(a) (i) Plot a graph of the student's results.

Join the points with straight lines.

(5)



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(ii) State two variables that should be controlled in this investigation.

(2)

1 .....

2 .....

(iii) Name the dependent variable in this investigation.

(1)

.....

5. Question 4 · part a.i · 5 marks · Working and answer

Working and answer

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6. Question 3 · part b.i · 5 marks

June 2021 Paper 1

Further practice · Attempt part b.i only.

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3 Enzymes are protein molecules.

(a) (i) Which sub-units form enzymes?

(1)

- ☒ **A** amino acids
- ☒ **B** fatty acids
- ☒ **C** glucose
- ☒ **D** glycerol

(ii) Describe how a solution can be tested to see if it contains protein.

(3)

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6. Question 3 · part b.i · 5 marks

June 2021 Paper 1

Further practice · Attempt part b.i only.

(b) A scientist investigates the effect of temperature on two different enzymes, A and B.

Enzyme A and enzyme B both digest proteins.

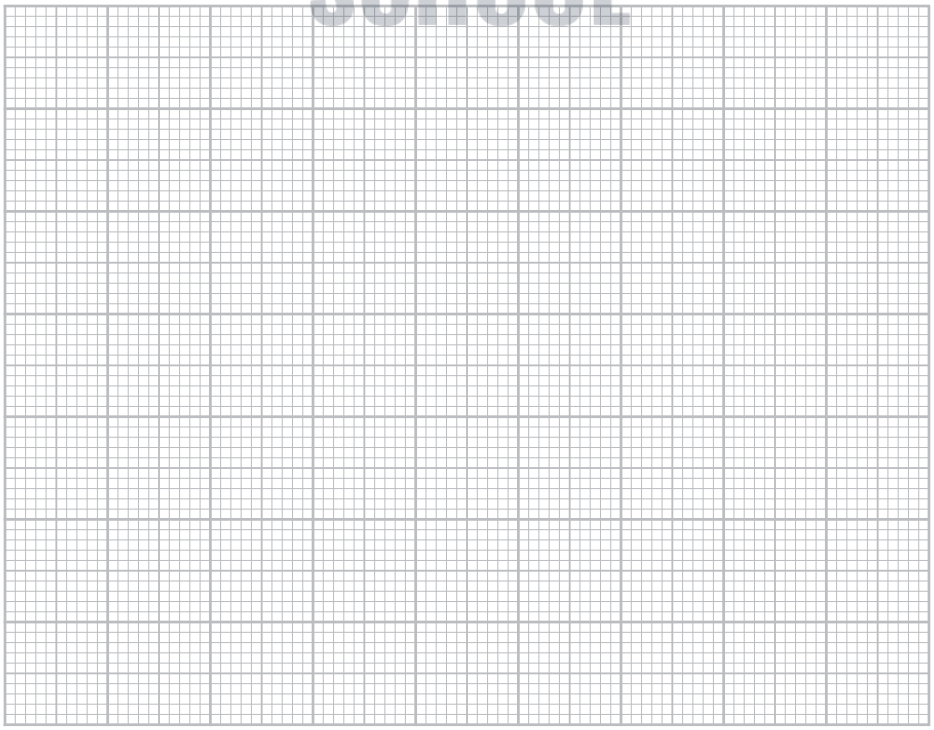
The results of the scientist's investigation are shown in the table.

Temperature in °C	Rate of protein digestion in arbitrary units	
	Enzyme A	Enzyme B
10	0	0
20	5	10
30	5	40
40	20	80
50	60	140
60	80	200
70	20	280
80	0	320
90	0	80

(i) Plot the scientist's results on the grid.

Draw the best fit curve for each enzyme.

(5)



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6. Question 3 · part b.i · 5 marks

June 2021 Paper 1

Further practice · Attempt part b.i only.

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(ii) Describe the effects on enzyme B of increasing the temperature from 80°C to 90°C.

(3)

(iii) Describe the differences between the rate of protein digestion shown by enzyme A and enzyme B.

Refer to the graph in your answer.

(3)

(Total for Question 3 = 15 marks)

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6. Question 3 · part b.i · 5 marks · Working and answer

Working and answer

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7. Question 2 · part b.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

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2 A student investigates the effect of a competitive inhibitor on the action of the enzyme phosphatase.

This is the reaction catalysed by the enzyme.

phenolphthalein phosphate ⟶ phenolphthalein + phosphate

Phenolphthalein turns pink when an alkali is added.

This colour can be measured using a colorimeter. The more intense the colour, the higher the reading on the colorimeter.

This is the student's method.

- prepare six different concentrations of phenolphthalein phosphate solution
- place 5.0 cm^3 of each solution into a different test tube
- add 1.0 cm^3 of a competitive inhibitor and 1.0 cm^3 of the enzyme phosphatase into each test tube
- wait 30 minutes, then add 1.0 cm^3 of an alkali to each tube
- measure the intensity of the pink colour using a colorimeter

(a) (i) Give a suitable control for this investigation.

(2)

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(ii) State three variables that should be kept constant in this investigation.

(3)

1 .....

2 .....

3 .....

7. Question 2 · part b.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

(b) The table shows the student's results.

Concentration of phenolphthalein phosphate solution in arbitrary units	Intensity of pink colour in arbitrary units
0.05	0.20
0.10	0.30
0.20	0.47
0.40	0.65
0.60	0.79
0.80	0.91

(i) Plot the results of the investigation. **(3)**

(ii) Draw a curve of best fit. **(1)**

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P 7 0 9 5 0 A 0 6 2 4

7. Question 2 · part b.i · 3 marks · Working and answer

Working and answer

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8. Question 5 · part b.i · 4 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

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5 Some athletes train at high altitudes to improve their performance.

Performance improves if more energy is released from aerobic respiration.

(a) (i) Complete the balanced symbol equation for aerobic respiration.

(2)

..... + $6O_2$ → ..... + $6CO_2$

(ii) Aerobic respiration produces ATP.

Describe how energy is released from ATP during exercise.

(2)

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8. Question 5 · part b.i · 4 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

(b) The table shows the percentage change in mass of haemoglobin in the blood after different periods of time training at altitude.

Period of time in hours	Percentage change in mass of haemoglobin
0	0
100	0.7
200	2.4
300	4.1
400	5.9
500	6.5
600	7.2

(i) Plot the results on the grid. **(4)**

(ii) Draw the curve of best fit. **(1)**

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8. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

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9. Question 5 · part b.i · 6 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

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5 A student uses this method to investigate the effect of exercise on the production of sweat.

- place a person in a room which has a temperature of 20 °C
- ask the person to remain inactive for 5 minutes
- measure the amount of sweat produced by a 25 cm^2 area of skin
- ask the person to exercise and measure the amount of sweat produced
- ask the person to rest for 10 minutes

Repeat the method at room temperatures of 22 °C, 24 °C, 26 °C, 28 °C and 30 °C

(a) (i) Explain why the person rests for 10 minutes before the method is repeated.

(2)

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(ii) State three variables that should be controlled.

(3)

1 .....

2 .....

3 .....

9. Question 5 · part b.i · 6 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

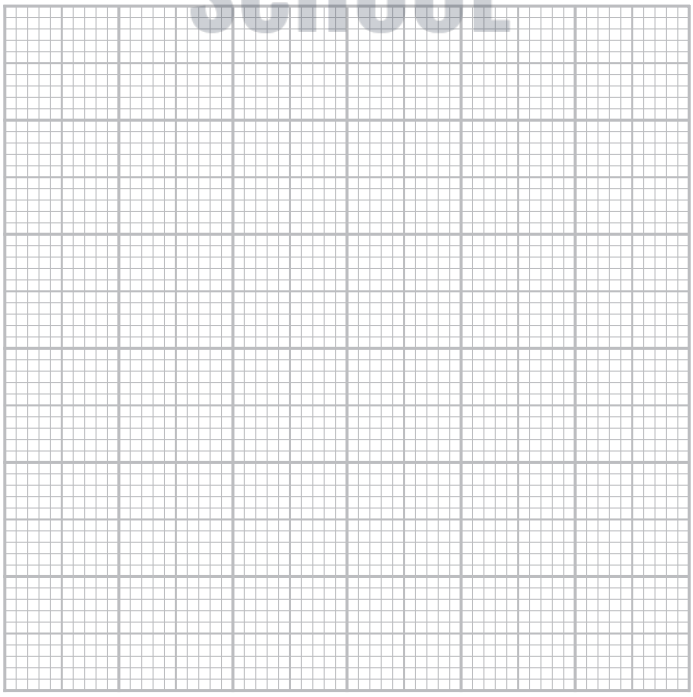
(b) The table shows the student's results.

Room temperature in °C	Amount of sweat produced by 25 cm^2 area of skin in arbitrary units	
	While inactive	During exercise
20	1	2
22	2	6
24	2	8
26	2	11
28	3	14
30	4	22

(i) Plot line graphs of the student's results.

Join the points with straight lines.

(6)



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P 7 2 5 8 6 A 0 1 2 2 0

9. Question 5 · part b.i · 6 marks

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Further practice · Attempt part b.i only.

(ii) Explain the differences between the two sets of results.

(5)

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(Total for Question 5 = 16 marks)

9. Question 5 · part b.i · 6 marks · Working and answer

Working and answer

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10. Question 7 · part a.i · 4 marks

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Further practice · Attempt part a.i only.

7 A method that shows whether a person's kidneys are working correctly is to measure the volume of filtrate produced by the kidneys in one minute. This is known as the glomerular filtration rate (GFR).

A person with healthy kidneys has a GFR of over 100 cm^3 per minute.

The GFR of a person whose kidneys were not working correctly was measured once a year for nine years.

The table shows the results.

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019
GFR in cm^3 per minute	80	75	79	79	67	60	50	45	35

(a) (i) Plot a line graph of the data on the grid.

(4)



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10. Question 7 · part a.i · 4 marks

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Further practice · Attempt part a.i only.

(ii) Explain why it is not possible to determine from the data when the person's kidneys stopped working correctly.

(2)

(iii) A person needs dialysis if the GFR falls to 15 cm^3 per minute.

Use the data to estimate when the person will probably need to start dialysis.

(2)

(iv) Discuss the accuracy of this estimate.

(3)

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10. Question 7 · part a.i · 4 marks · Working and answer

Working and answer

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11. Question 4 · part a.i · 3 marks

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Further practice · Attempt part a.i only.

4 (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

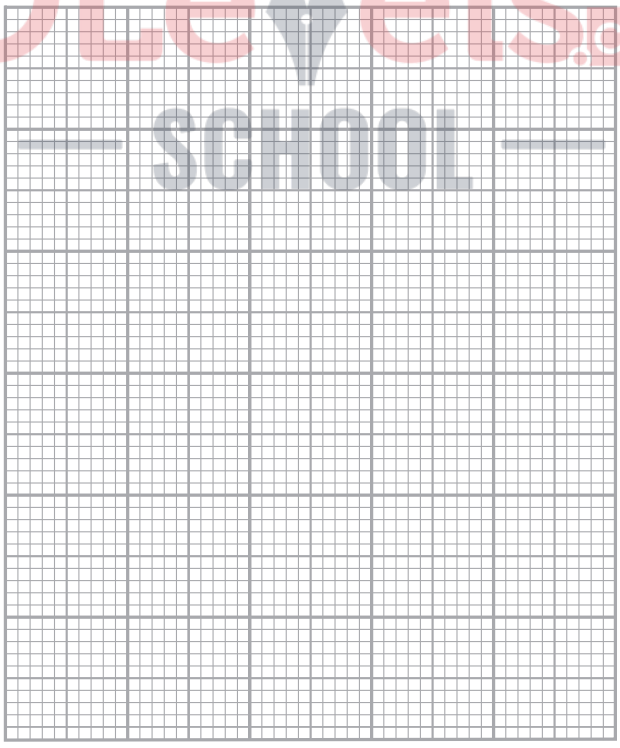
pH	Time taken to digest starch in minutes
2	60
4	45
6	20
8	35
10	55

(i) Plot a graph using the data in the table.

(3)

(ii) Draw a curve of best fit.

(1)



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11. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

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12. Question 2 · part a.i · 3 marks

November 2024 Paper 1

Further practice · Attempt part a.i only.

2 A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

Time in minutes	pH of solution
1	6.8
2	6.4
3	6.0
4	5.8
5	5.6
6	5.2
7	5.0
8	4.8
9	4.6
10	4.6

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12. Question 2 · part a.i · 3 marks

November 2024 Paper 1

Further practice · Attempt part a.i only.

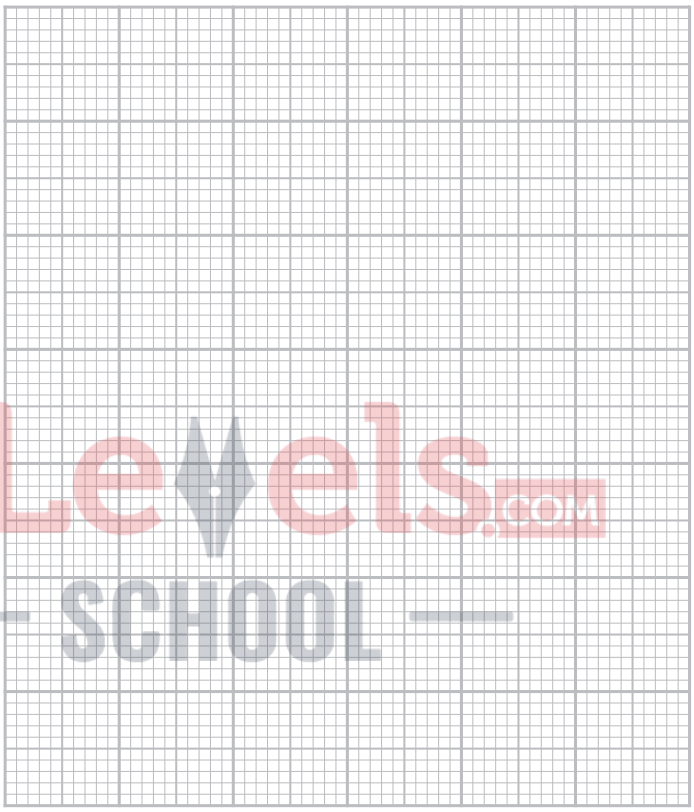
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(a) (i) Plot a graph of the student's results.

(3)



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(ii) Draw a line of best fit.

(1)

(iii) Give the trend shown by the graph.

(1)

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12. Question 2 · part a.i · 3 marks · Working and answer

Working and answer

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13. Question 4 · part a.i · 4 marks

June 2025 Paper 1R

Further practice · Attempt part a.i only.

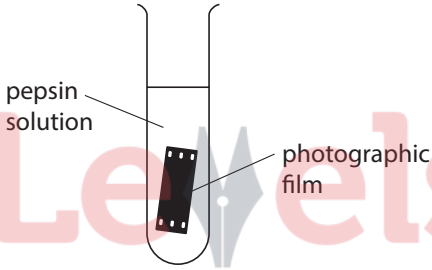
4 Photographic film is made of clear plastic, coated in a protein. The protein contains silver compounds which give the film its black colour.

Pepsin is an enzyme that digests protein.

A student investigates how the concentration of pepsin affects the time taken to digest the protein in a piece of photographic film.

This is the student's method.

- place a piece of photographic film into a test tube
- add a 1% pepsin solution
- measure the time for the film to become clear
- repeat the investigation using other concentrations of pepsin



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The table shows the student's results.

Percentage pepsin concentration (%)	Time taken for film to become clear in seconds
1	112
2	102
3	93
4	84
5	84
6	84

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13. Question 4 · part a.i · 4 marks

June 2025 Paper 1R

Further practice · Attempt part a.i only.

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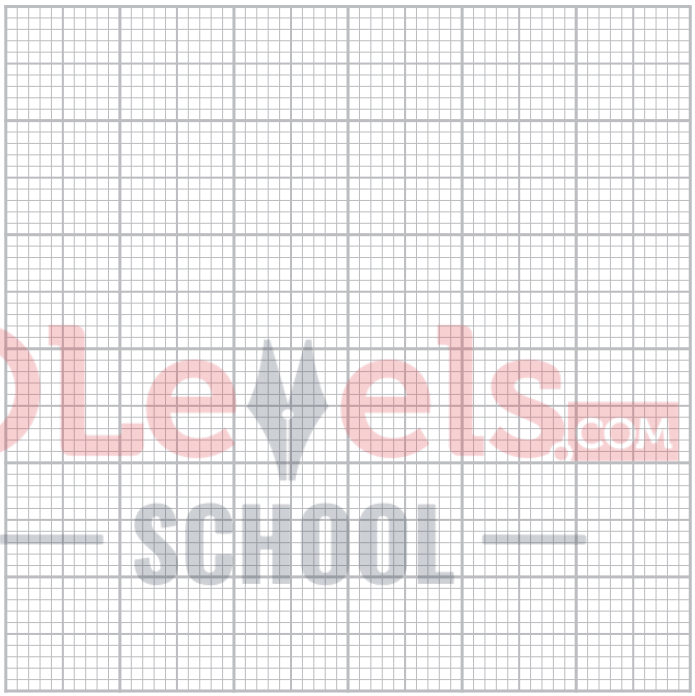
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(a) (i) Plot a graph of the results on the grid.

(4)

(ii) Draw a curve of best fit.

(1)



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(b) (i) Explain why the photographic film becomes clear.

(3)

13. Question 4 · part a.i · 4 marks · Working and answer

Working and answer

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14. Question 4 · part c.i · 5 marks

June 2025 Paper 1

Further practice · Attempt part c.i only.

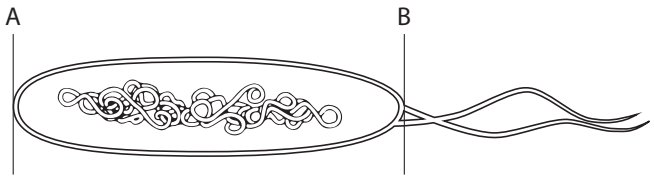
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4 (a) State what is meant by the term **pathogen**.

(1)

.....

(b) The diagram shows a pathogenic bacterium.



(i) Label **three** structures shown on the diagram.

(3)

(ii) The actual length of the bacterium between points A and B is 2.0 µm.

Calculate the magnification of the diagram.

[1 mm = 1000 µm]

(3)

magnification = × .....

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14. Question 4 · part c.i · 5 marks

June 2025 Paper 1

Further practice · Attempt part c.i only.

(c) The effectiveness of a vaccine in preventing disease was studied over six months.

Two different groups of people were used. One group of people were aged between 45 and 64 years and the second group were aged over 65 years.

The results are shown in the table.

	Percentage effectiveness against disease (%)	
Time since vaccination in months	**45 to 64 years**	**over 65 years**
1	90	80
2	85	78
3	80	75
4	68	58
5	60	50
6	50	40

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14. Question 4 · part c.i · 5 marks

June 2025 Paper 1

Further practice · Attempt part c.i only.

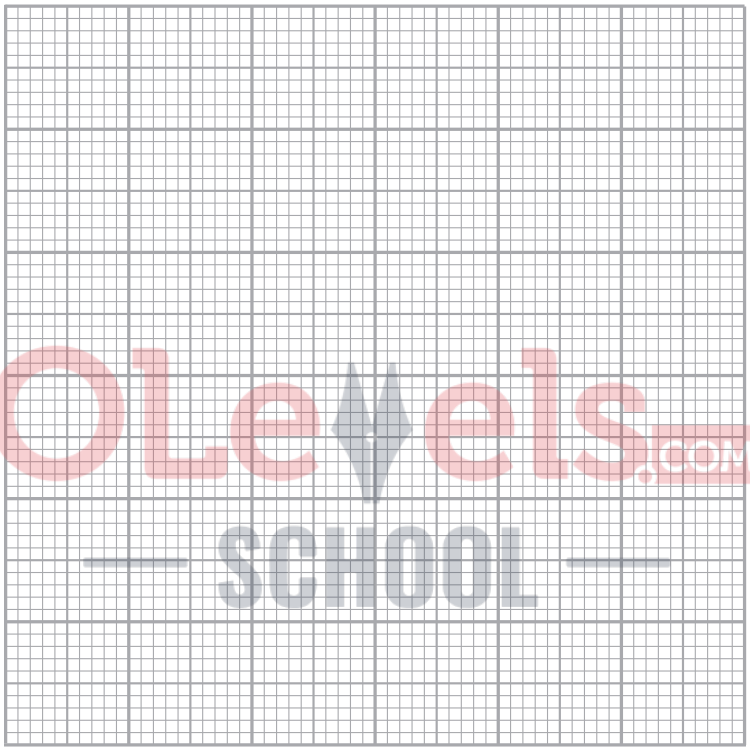
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(i) Plot a graph of the results. Join the points with straight lines.

(5)



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(ii) Describe the trends shown by this graph.

(3)

(Total for Question 4 = 15 marks)

14. Question 4 · part c.i · 5 marks · Working and answer

Working and answer

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